Roll No. [ ] Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem-4)

PHARMACEUTICAL ORGANIC CHEMISTRY-III

Subject Code : BP-401T

M.Code : 75843

Date of Examination : 14-06-2023

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Write short notes on :**

   a) Define Heterocyclic compounds with examples.

   b) Define the term chiral molecule. What is necessary condition for optical activity?

   c) Explain configuration vs conformation.

   d) Draw all possible conformation of Ethane and Cyclohexane.

   e) Write the structure of meso tartaric acid in Fischer projection. Comment on its optical activity.

   f) Write the resonance structures of Pyridine and Pyrimidine.

   g) Write chemical structure with numbering of Imidazole.

   h) Write basic structure and uses of Purine.

   i) Write synthesis and medicinal use of Sodium Borohydride.

   j) Write the structure and uses of Sodium Borohydride.

SECTION-B

2. Discuss synthesis of Fisher-Indole synthesis and Skarup synthesis of Quinoline.

3. Write explanatory note on stereochemistry of biphenyl compounds and asymmetric synthesis.

4. Write, detailed note on Oppenauer-oxidation and Dakin reaction.

SECTION-C

5. Explain the relative aromaticity of thiophene in contrast to furan and pyrrole.

6. Illustrate using example DL system and RS system of nomenclature of optical isomers.

7. Discuss stereospecific or stereoselective reactions using one suitable example.

8. Discuss mechanism and synthetic importance of Wolff Kishner reduction.

9. Differentiate between Enantiomerism and Diastereomerism using suitable examples.

10. Write synthesis and uses of Acridine. Discuss basicity of pyridine.

11. Explain mechanism and synthetic importance of Claisen-Schmidt condensation.

12. Discuss racemic modification and resolution of racemic mixture.

13. Give, brief note on methods of determination of configuration of geometrical isomers.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

July-2023